Antonio Anna Mele

Quantum Information • Freie Universität Berlin • Google PhD Fellow

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Education

2021–Present **PhD in Quantum Information**, Freie Universität Berlin, Germany. *Advisor:* Jens Eisert.

Recipient of the **2025 Google PhD Fellowship in Quantum Computing**, awarded to *"outstanding graduate students doing exceptional and innovative research"*.

2019–2021 **MSc in Physics** (Double Degree), University of Trento – SISSA, Italy. Graduated with 110/110 *cum laude* (highest possible grade); maximum average exam score.

2016–2019 **BSc in Physics**, University of Pisa, Italy. Graduated with 110/110 *cum laude* (highest possible grade).

Research Positions

Summer 2026 **Graduate Student Researcher**, Google Quantum AI, Los Angeles, CA, USA. *Mentor:* Jarrod McClean.

Summer 2024 **Graduate Student Researcher**, Los Alamos National Laboratory, NM, USA. *Mentor:* Marco Cerezo.

Publications (Google Scholar, $> 1{,}600$ citations, h-index 17; selected papers marked with $\star$)

1. $\star$ **A. A. Mele**, L. Bittel. *Optimal learning of quantum channels in diamond distance*. *Nature Physics*, accepted for publication (2026). arXiv:2512.10214.
2. $\star$ **A. A. Mele**, A. Angrisani, S. Ghosh, S. Khatri, J. Eisert, D. S. França, Y. Quek. *Noise-induced shallow circuits and the absence of barren plateaus*. *Nature Physics* 22, 751–756 (2026).
3. $\star$ F. A. Mele, **A. A. Mele**, L. Bittel, J. Eisert, V. Giovannetti, L. Lami, L. Leone, S. F. E. Oliviero. *Learning quantum states of continuous-variable systems*. *Nature Physics* 21, 2002–2008 (2025).
4. $\star$ **A. A. Mele***, F. A. Mele*, J. R. McClean, T. E. O'Brien. *Towards verifiable quantum advantage with random circuits: Observables that survive concentration*. arXiv:2609.37890 (2026). *(* Equal contributors.)*
5. $\star$ **A. A. Mele**, Y. Herasymenko. *Efficient learning of quantum states prepared with few fermionic non-Gaussian gates*. *PRX Quantum* 6, 010319 (2025).
6. $\star$ **A. A. Mele**. *Introduction to Haar measure tools in quantum information: A beginner's tutorial*. *Quantum* 8, 1340 (2024).
7. $\star$ S. Chen, **A. A. Mele**†, F. A. Mele, J. Preskill. *When are bosonic Gaussian states classical to learn?* arXiv:2609.26705 (2026). *(† Authors listed alphabetically.)*
8. $\star$ T. Parella-Dilmé, J. Barberà-Rodríguez, S. F. E. Oliviero, **A. A. Mele**. *Strong unitary designs in optimal depth and space*. arXiv:2608.13491 (2026).
9. $\star$ L. Bittel, J. Eisert, L. Leone, **A. A. Mele**†, S. F. E. Oliviero. *A complete theory of the Clifford commutant*. *Quantum* 10, 2171 (2026). *(† Authors listed alphabetically.)*
10. $\star$ A. Angrisani, **A. A. Mele**, M. S. Rudolph, M. Cerezo, Z. Holmes. *Simulating quantum circuits with arbitrary local noise using Pauli propagation*. *PRX Quantum* 7, 020313 (2026).
11. $\star$ L. Bittel, **A. A. Mele**, J. Eisert, L. Leone. *Optimal trace-distance bounds for free-fermionic states: Testing and improved tomography*. *PRX Quantum* 6, 030341 (2025).
12. $\star$ L. Bittel, F. A. Mele, J. Eisert, **A. A. Mele**. *Energy-independent tomography of Gaussian states*. arXiv:2508.14979 (2025).
13. L. Bittel, J. Eisert, W. Gong, **A. A. Mele**†, L. Schatzki. *Adaptivity is all you need: Optimal stabilizer learning using just single-copy measurements*. arXiv:2610.02031 (2026). *(† Authors listed alphabetically.)*
14. M. Yaghubi Rad, R. Puig, S. Hassanandani, L. Coffman, J. Eisert, C. Bravo-Prieto, **A. A. Mele**. *Testing quantum Gaussianity with constant sample complexity*. arXiv:2609.40270 (2026).
15. C. Bravo-Prieto, W. Gong, **A. A. Mele**†. *Quantum memory advantage for quantum process tomography*. arXiv:2607.13476 (2026). *(† Authors listed alphabetically.)*

16. D. Miller, K. Levi, L. Postler, A. Steiner, L. Bittel, G. A. L. White, Y. Tang, **A. A. Mele**, et al. *Experimental measurement and a physical interpretation of quantum shadow enumerators*. *Physical Review Research* 8, 023318 (2026) Editor suggestion.
17. A. E. Deneris, P. Braccia, P. Bermejo, N. L. Diaz, **A. A. Mele**, M. Cerezo. *Analyzing the free states of one quantum resource theory as resource states of another*. *Advanced Quantum Technologies* 9, e00702 (2026).
18. M. Fanizza, V. Iyer, J. Lee, **A. A. Mele**†, F. A. Mele. *Efficient learning of bosonic Gaussian unitaries*. arXiv:2510.05531 (2025). *(† Authors listed alphabetically.)*
19. N. L. Diaz, **A. A. Mele**, P. Bermejo, P. Braccia, A. E. Deneris, M. Larocca, M. Cerezo. *A unified approach to quantum resource theories and a new class of free operations*. arXiv:2507.10851 (2025).
20. P. Bermejo, P. Braccia, **A. A. Mele**, N. L. Diaz, A. E. Deneris, M. Larocca, M. Cerezo. *Characterizing quantum resourcefulness via group-Fourier decompositions*. arXiv:2506.19696 (2025).
21. L. Bittel, F. A. Mele, **A. A. Mele**, S. Tirone, L. Lami. *Optimal estimates of trace distance between bosonic Gaussian states and applications to learning*. *Quantum* 9, 1769 (2025).
22. M. West, **A. A. Mele**, M. Larocca, M. Cerezo. *Real classical shadows*. *Journal of Physics A: Mathematical and Theoretical* 58, 245304 (2025).
23. J. Liu, F. Wilde, **A. A. Mele**, X. Jin, L. Jiang, J. Eisert. *Stochastic noise can be helpful for variational quantum algorithms*. *Physical Review A* 111, 052441 (2025).
24. L. Bittel, **A. A. Mele**, J. Eisert, L. Leone. *PAC-learning of free-fermionic states is NP-hard*. *Quantum* 9, 1665 (2025).
25. J. Denzler, **A. A. Mele**, E. Derbyshire, T. Guaita, J. Eisert. *Learning fermionic correlations by evolving with random translationally invariant Hamiltonians*. *Physical Review Letters* 133, 240604 (2024).
26. M. West, **A. A. Mele**, M. Larocca, M. Cerezo. *Random ensembles of symplectic and unitary states are indistinguishable*. arXiv:2409.16500 (2024).
27. J. J. Meyer, M. Mularski, E. Gil-Fuster, **A. A. Mele**, F. Arzani, A. Wilms, J. Eisert. *Exploiting symmetry in variational quantum machine learning*. *PRX Quantum* 4, 010328 (2023).
28. **A. A. Mele**, G. B. Mbeng, G. E. Santoro, M. Collura, P. Torta. *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian variational ansatz*. *Physical Review A* 106, L060401 (2022).

Teaching & Invited Lectures

05/2026 **Invited Lecturer**, EPFL Bernoulli Center, Lausanne, Switzerland. *Quantum Learning Theory for Bosonic and Fermionic Systems* (1.5-hour blackboard tutorial), delivered as part of the program *Intersection of Quantum Learning Theory, Quantum Sensing, and Quantum Verification*. [program] [notes] [video]

01/2026 **Invited Lecturer**, 2026 QISCA Winter School. Lecture on *Learning t-doped Fermionic and Bosonic Gaussian States*, as part of *Quantum Learning Theory for Bosonic and Fermionic Systems*. [course] [notes] [video] [fermionic paper] [bosonic paper]

10/2025 **Invited Lecturer**, AIMS Workshop and School on *The Theory of Quantum Learning Algorithms*, Cape Town, South Africa. Four-hour tutorial on *Haar Integration Tools in Quantum Information*. [program] [Haar tutorial]

08/2025 **Invited Lecturer**, QMATH Masterclass, University of Copenhagen (6 lectures). *Representation Theory in Quantum Information Science*. 3 lectures on representation theory (original lecture notes prepared for the masterclass: PDF); 3 lectures on Haar measure and Clifford commutant methods, based on *Quantum* (2024) and arXiv:2504.12263. (Masterclass link)

2025 **MSc Thesis Mentor**, Diego Polimeni. *Rigorous End-to-End Fault-Tolerant Quantum Computation with Minimal Assumptions*.

2022–2023 **Teaching Assistant**, *Quantum Information Theory*, Freie Universität Berlin. Led exercise sessions and tutorials; [course materials].

Academic Service

Program Committees — PC member for QIP 2025, QTML 2025, QCTiP 2026, QTML 2026 and (forthcoming) QIP 2027.

Refereeing — Referee for QIP, TQC, QTML, and QCTiP over multiple years, as well as for physics journals.

Research visits

03/2026 **California Institute of Technology (Caltech)**, Pasadena, USA. Hosted at IQIM by Hsin-Yuan (Robert) Huang and John Preskill.

02/2026 **Massachusetts Institute of Technology (MIT)**, Cambridge, USA. Hosted at CTP by Soonwon Choi.

02/2026 **Harvard University**, Cambridge, USA. Hosted at HQI by Susanne Yelin.

05/2025 **Perimeter Institute for Theoretical Physics**, Waterloo, Canada. Hosted by Farzin Salek.

Awards & Fellowships

2026 **Anthropic AI for Science Program**. Selected by Anthropic for research support, with an award of US$20,000 in usage credits.

2027-Present **Simons Institute–Jane Street Circles Program**. Collaborative quantum algorithms project selected for funding in the Circles program co-hosted by the Simons Institute for the Theory of Computing and Jane Street, supporting four weeks of in-person research collaboration over two years.

2026 **Federico Tonielli Award**, conferred at a ceremony at the Scuola Normale Superiore in Pisa.

2025-Present **Google PhD Fellowship in Quantum Computing** (US$70,000/year in research support). Awarded by Google to "recognize outstanding graduate students doing exceptional and innovative research in areas relevant to computer science and related fields."

2019 **University of Trento–SISSA MSc Scholarship**. One of seven recipients, selected through a competitive examination in physics and mathematics. Included financial support and access to advanced PhD-level courses at International School for Advanced Studies (SISSA).

Talks

Selected conference contributions

01/2026 *A complete theory of the Clifford commutant and applications*. QIP 2026.

01/2026 *Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators*. QIP 2026.

09/2025 *Pauli Propagation: A computational framework for simulating quantum systems*. QTML 2025.

09/2025 *A complete theory for the Clifford commutant*. TQC 2025.

02/2025 *Learning and testing quantum states of fermionic systems*. QIP 2025.

02/2025 *Tomography of bosonic systems and optimal estimates of the trace distance between Gaussian states*. QIP 2025.

04/2025 *Tomography of bosonic systems and optimal estimates of the trace distance between Gaussian states*. QCTiP 2025.

09/2024 *Learning quantum states of continuous variable systems*. QTML 2024.

09/2024 *Learning quantum states of continuous variable systems*. TQC 2024.

09/2024 *Noise-induced shallow circuits and absence of barren plateaus*. TQC 2024.

08/2024 *Learning quantum states of continuous variable systems*. AQIS 2024.

08/2024 *Learning and testing possibly magical fermions*. AQIS 2024.

08/2024 *Noise-induced shallow circuits and absence of barren plateaus*. AQIS 2024.

11/2022 *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian Variational Ansatz*. QTML 2022 (extended talk).

09/2022 *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian Variational Ansatz*. DPG 2022.

09/2021 *Exploiting symmetry in variational quantum machine learning*. DPG 2021.

Selected invited talks

10/2026 *Learning and computation in quantum systems*, Federico Tonielli Award ceremony, Scuola Normale Superiore, Pisa (16 October 2026). [slides]

10/2026 Invited Speaker (forthcoming), *Quantum Innovators Theory Workshop 2026*, Institute for Quantum Computing, University of Waterloo. Workshop bringing together early-career researchers making an impact on quantum information research.

05/2026 *Quantum Learning Theory for Bosonic and Fermionic Systems*, EPFL Bernoulli Center (program on *Intersection of Quantum Learning Theory, Quantum Sensing, and Quantum Verification*). (link)

03/2026 *Optimal learning of quantum channels in diamond distance*, Caltech IQIM Seminar.

02/2026 *Optimal learning of quantum channels in diamond distance*, MIT Quantum Seminar.

02/2026 *Optimal learning of quantum channels in diamond distance*, Harvard Quantum Seminar.

09/2025 *A complete theory for the Clifford commutant*, Los Alamos National Laboratory.

08/2025 *Invited lecturer at the QMATH Masterclass on Representation Theory in Quantum Information Science* (masterclass link), University of Copenhagen.

08/2025 **TEDx Policoro**. *Quantum Computer: una possibile rivoluzione*, with Francesco A. Mele. [video] [event]

07/2025 *Effect of noise in typical quantum circuits*, *Understanding Quantum Machine Learning* workshop, Galileo Galilei Institute, Florence, Italy.

05/2025 *Learning and testing quantum states of fermionic systems*, Perimeter Institute for Theoretical Physics, Waterloo, Canada.

05/2025 *Effect of noise in typical quantum circuits*. The Higgs Centre for Theoretical Physics, Edinburgh, UK.

04/2025 *Effect of noise in typical quantum circuits*. Scuola Normale Superiore (SNS), Pisa, Italy.

09/2024 *Noise-induced shallow circuits and absence of barren plateaus*. Quantinuum.

04/2024 *Noise-induced shallow circuits and absence of barren plateaus*. Centre for Quantum Technologies, NUS.

04/2024 *Noise-induced shallow circuits and absence of barren plateaus*. BraQIIIT Lab, IIIT-Delhi.

Schools and Workshops

08/2025 *QMATH Masterclass on Representation Theory in Quantum Information Science*, University of Copenhagen, invited lecturer.

07/2025 *Workshop on the Clifford commutant*, Cologne. Organized to discuss our work.

07/2025 *Understanding Quantum Machine Learning*, Galileo Galilei Institute, Florence, invited speaker.

05/2025 *Entanglement, Information and Complexity in Quantum Systems*, Edinburgh, invited speaker.

06–08/2024 *Quantum Computing Summer School*, Los Alamos National Laboratory (acceptance ratio $< 3\%$).

2022–2024 QMATH Masterclasses: Entropy Inequalities (2022), Quantum Learning Theory (2023), and Quantum Simulation (2024), University of Copenhagen.

07/2023 *PCMI Graduate Summer School in Quantum Computation*, IAS/Park City.

Earlier Achievements

2015 **Italian Informatics Olympiad**. National finalist.

2015 **IoStudio Award** (MIUR & Samsung). Winner for developing an Android application for my high school.

2014–2016 **Elected Student Body Representative**, Liceo Scientifico "Enrico Fermi" (Policoro, Italy) — two consecutive years; public speaking and organizing high school workshops and student activities.

MSc thesis *Quantum Approximate Optimization Algorithm for Quantum Many-Body States Preparation*. Advisors: Mario Collura; Philipp H.J. Hauke.

BSc thesis *How to break down quantum noise in a gravitational waves detector*. Advisor: Giancarlo Cella.

Theory, AI & Scientific Computing

Theory Quantum information, quantum algorithms and complexity; broad mathematical background, with particular expertise in representation theory, probability theory, matrix analysis, and random matrix theory, and experience rapidly learning and applying new mathematical tools.

AI & ML	Learning theory and statistical inference, with particular emphasis on quantum learning; large language models for mathematical reasoning, scientific programming, and AI-assisted research.
Computing	Python, Mathematica, C/C++; scientific computing and numerical linear algebra; algorithm implementation and numerical benchmarking; quantum-circuit and many-body simulation; Cirq, Qiskit, PennyLane.